

EXPERIMENT 3

TITLE

Breadth First Search (BFS)

AIM

To write a Python program to implement the **Breadth First Search (BFS)** algorithm for graph traversal.

OBJECTIVE

- To understand the Breadth First Search algorithm.
- To traverse a graph level by level.
- To visit all vertices using a queue.
- To demonstrate uninformed search in Artificial Intelligence.

ALGORITHM

Step 1: Create the graph.

Step 2: Create an empty queue.

Step 3: Insert the starting node into the queue.

Step 4: Mark the node as visited.

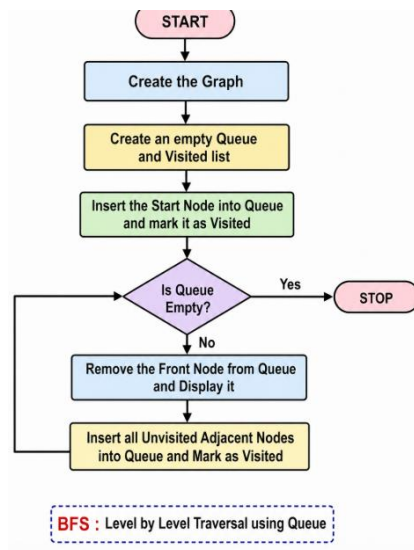
Step 5: Remove a node from the queue.

Step 6: Display the node.

Step 7: Insert all unvisited adjacent nodes into the queue.

Step 8: Repeat Steps 5–7 until the queue becomes empty.

FLOWCHART



PYTHON CODE

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

visited = []
queue = []

def bfs(node):
    visited.append(node)
    queue.append(node)

    while queue:
        s = queue.pop(0)
        print(s, end=" ")

        for neighbour in graph[s]:
            if neighbour not in visited:
                visited.append(neighbour)
                queue.append(neighbour)

bfs('A')
```

SAMPLE OUTPUT

A B C D E F

CONCLUSION

The Breadth First Search algorithm was implemented successfully. It visits all nodes level by level using a queue and guarantees finding the shortest path in an unweighted graph.